result in very realistic approximations of global illumination at a much lower computational cost than standard ray tracing.

VR Direct
Latency is a near killer when it comes to Virtual Reality. In fact it is one of the prime causes of motion sickness associated with any Virtual Reality equipment. To put it simply, the disconnect or lag between you moving your head and the scene being rendered in front of your eyes causes your brain to go "wut?" and behave erratically. VR Direct is an attempt by NVIDIA to bring down this latency in order to truly take VR mainstream; the first and most important step of this would be to up the resolution for current VR implementations to at least 4K. When the screen is so close to your eye, one megapixel per eye is just not enough. For this the latency between SLI (of course you'll need two of these powerplants) should be zero or near zero.

Using things like MFAA and Asynchronous Ward, VR Direct has managed to bring down latency in the compute stack to 25 ms. The other interesting feature of VR Direct is Auto Stereo which is supposed to allow you to play games that weren't designed for VR in VR!

Architecture
Being the flagship the GTX 980 has every single component in a functional state and can be said to be the full implementation of the GM204 chip. Needless to say, lower SKUs will come with some of these components either soft-disabled or lasered out. The GTX 970 which also launched with the 980 comes with 13 SMMs and soon enough within a month we'll see the GTX 960 launch with about 10-11 SMMs and perhaps a GTX 960 Ti could release a few months later with 11-12 SMMs going by NVIDIA's history. We don't have any details at the moment and can only speculate on these SKUs.

As for memory, the GM204 has a 256-bit memory interface with 7 Gbps GDDR5 memory that results in a 224 GB/sec memory bandwidth which when compared to the GTX 780 Ti's 336 GB/s is quite less. However, with the new architecture comes improvement in caching and compression which has reportedly resulted in about 25% less bytes per frame which means it can effectively carry more data. So the 7 Gbps on Maxwell translates to 9.3 Gbps on Kepler.

Build
At first glance, the only thing that tells the card apart from its flagship predecessors would the engraved "GTX 980" that appears on the housing towards the mounting bracket. Flipping the card around shows us the biggest change with the reference build – the backplate. Yes, NVIDIA has taken it up a notch and covered the rear as to protect the SMD components. NVIDIA has made a small portion of the backplate near the power connectors removable which when pulled out opens up a channel for air to flow from the crevice into the fan of the SLI partner card.

The GTX 980 chucks one of the DVI ports and in its place gives you two more DisplayPorts. You can run four of these ports at any given time or three can be used when your monitor's are G-sync capable. Also, the vents on the rear seems to have been widened to allow for more airflow. The new design has a honeycomb like structure but with more truss elements for added rigidity. And the last thing to note is that the GTX 980 has two 6-pin power connectors because of the improved power efficiency.

Performance
The GeForce GTX 980 seems to perform better than every other card that we've tested so far on medium settings. On Ultra, it performs at the same level that factory overclocked editions of the GTX 780 Ti does. Computational benchmarks puts the Maxwell architecture in league with the AMD GPUs who'd ruled this arena for quite a few generations. So overall, this card scores better than every other card that has graced the test center with its presence.

Maxwell, does indeed deliver given that it uses less CUDA cores than Kepler to produce the same if not better results. At the end of the day, Maxwell uses lesser amount of transistors, occupies lesser area, consumer lesser wattage to produce better results overall in gaming as well as computational benchmarks.

Verdict
However, a launch price of Rs.46,000 is indeed surprising and we hear that price cuts will be announced across the entire range within this quarter. This will make this card a really sweet deal to let pass by. NVIDIA aims to target majority of users who're yet to buy a 700 series GPU. A recent survey showed that 68 percent of steam gamers have a 600 series or older GPU and with the lucrative pricing that the GTX 980 and the GTX 970 have launched this september tells us that these gamers are going to find it hard pressed to not upgrade.

Written with excerpts from Siddharth Parwatay. To understand the new features in detail check out these online articles. http://dgit.in/gfgtx980 and http://dgit.in/GTX980

Mithun Mohandas

Performance................ 90
Value 75
Build 60

Specifications
Chipset: GM204; **Base clock:** 1126MHz; **Memory clock:** 1750MHz; **CUDA cores:** 2048; **Texture Units:** 128; **ROPs:** 64; **Manufacturing process:** 28nm; PCIe 3.0, 4096 x 2160 digital resolution support, 4 GB Memory; **DirectX support:** 12; **OpenGL support:** 4.4; **Power Connectors:** 6Pin + 6Pin; **TDP:** 165W; **Dimensions (LxHxD):** 267mm x 112mm x 53mm; **Warranty:** NA

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